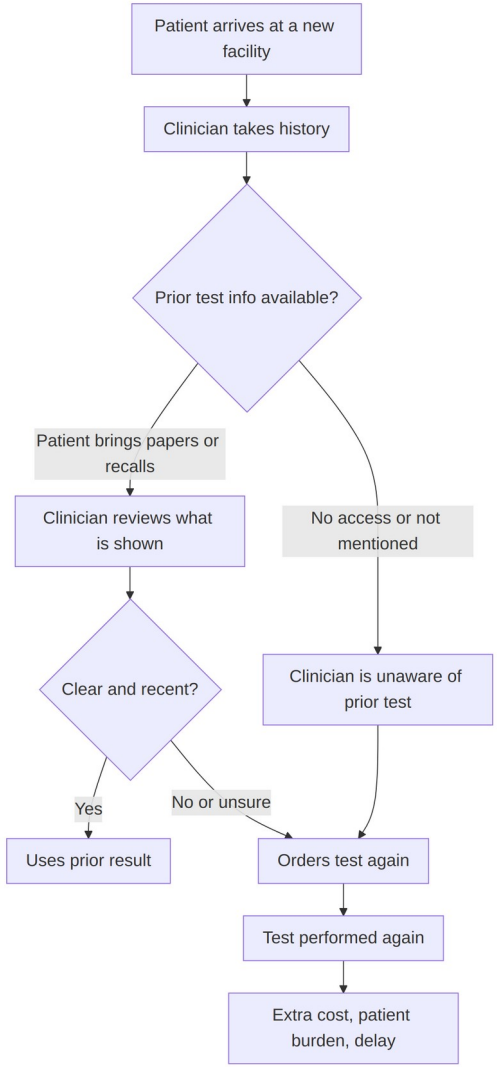
DupeCheck: Project Flow

Field	Detail
Purpose	One place that shows how DupeCheck works end to end: the clinical process today, the process with DupeCheck, the decision logic, the system flow, the data flow and the delivery flow
Related	DupeCheck BRD, DupeCheck PRD
Status	Draft v0.1. Flows marked **[Validate]** depend on EHR and data source decisions still to be confirmed

1. Flow at a glance

1. **Clinical flow today:** how a repeat test slips through.
2. **Clinical flow with DupeCheck:** where the alert enters.
3. **Decision logic:** exactly how a duplicate is judged.
4. **System sequence:** which components talk to each other.
5. **Data flow:** where the data comes from and goes.
6. **Override and governance flow:** how rules stay safe and useful.
7. **Delivery flow:** how the project moves from discovery to scale.

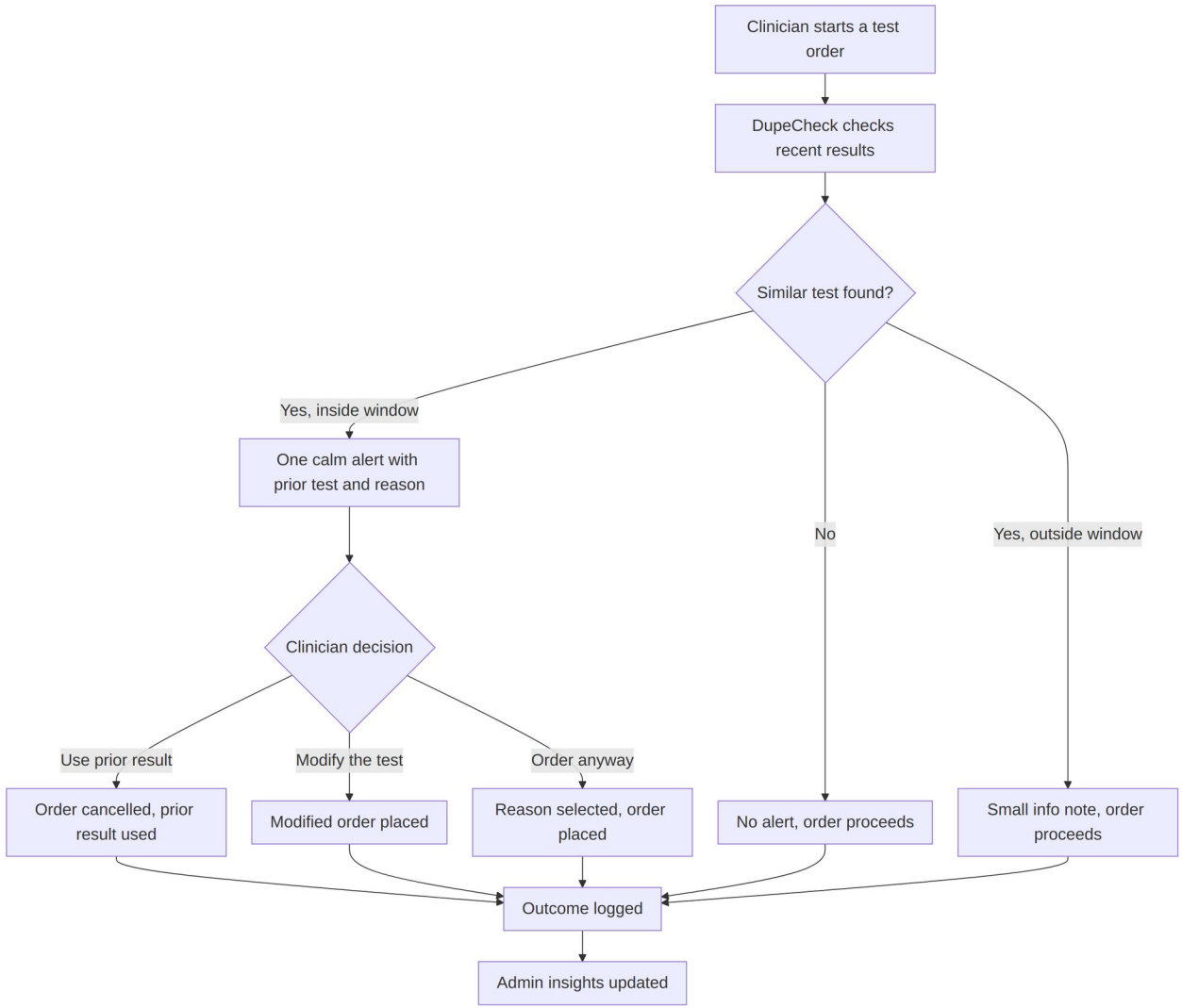
2. Current state: how a repeat test happens today



Pain points: reliance on the patient's memory or paperwork, manual record requests, no safeguard at the moment of ordering, no measurement of repeats.

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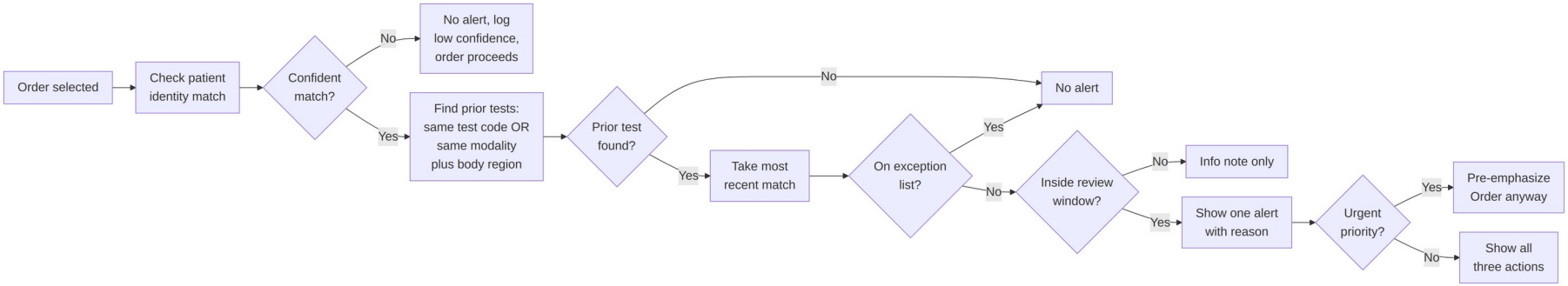
3. Future state: clinical flow with DupeCheck



What changes: the check runs automatically at the moment of the decision, the alert explains itself, the clinician stays in control, and every outcome is measured.

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4. Decision logic: how a duplicate is judged

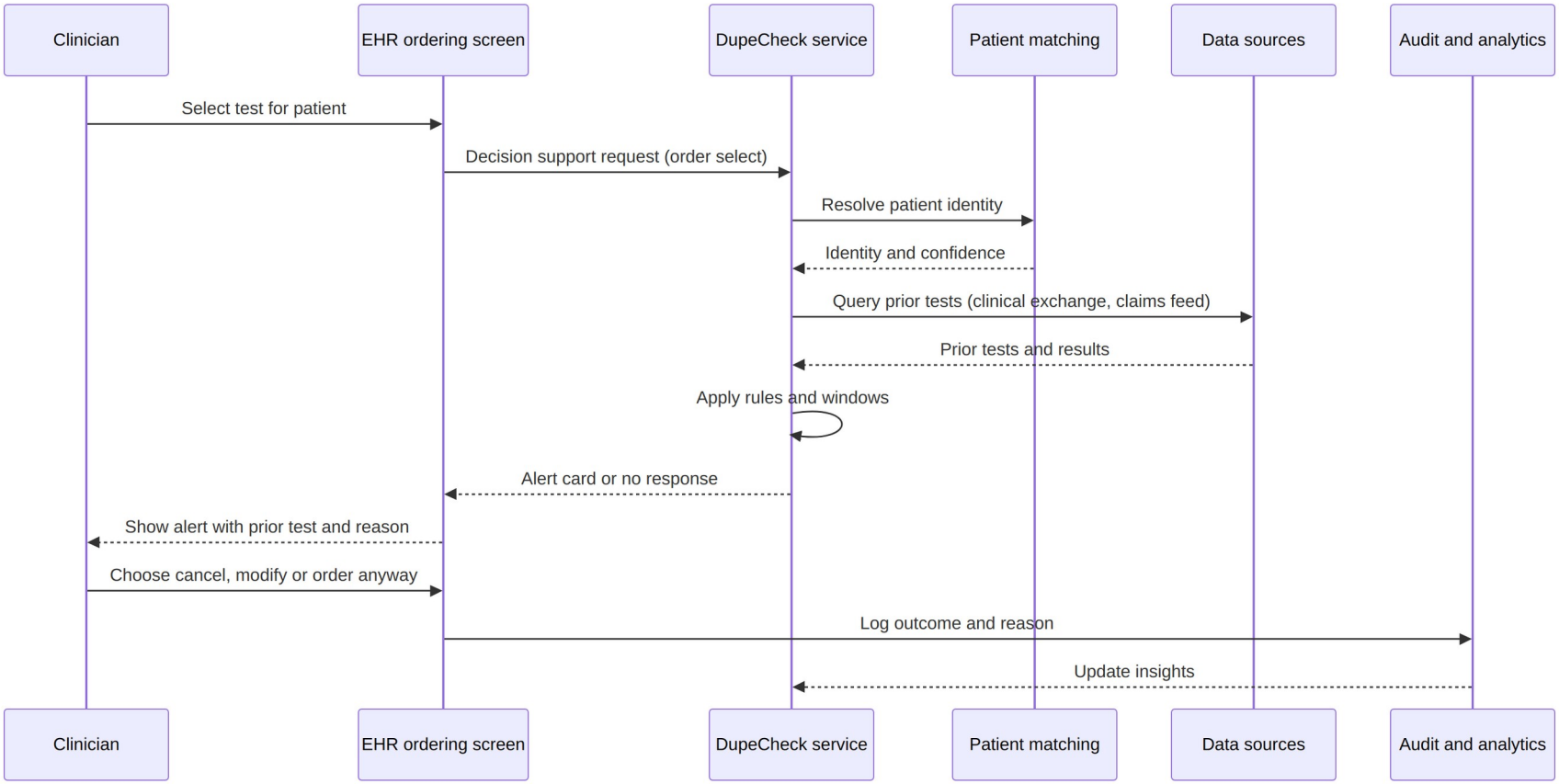


Key rules

- **Identity first.** A wrong patient match would cause a wrong alert, so low confidence means no alert.
- **Most recent match wins.** One alert per order, never a stack.
- **Never block.** No branch prevents an order from being placed.
- **Windows are illustrative [Validate]:** CT and MRI 90 days, X-ray 30, HbA1c 60, lipid panel 30, CBC 14, echocardiogram 180.

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5. System sequence (target architecture) [Validate]



Notes

- The integration pattern (a decision support hook inside the EHR) is a proposal. Confirm what the target EHR supports.
- If any data source is slow or unavailable, DupeCheck returns nothing and ordering continues unaffected.
- Target response time is under 2 seconds.

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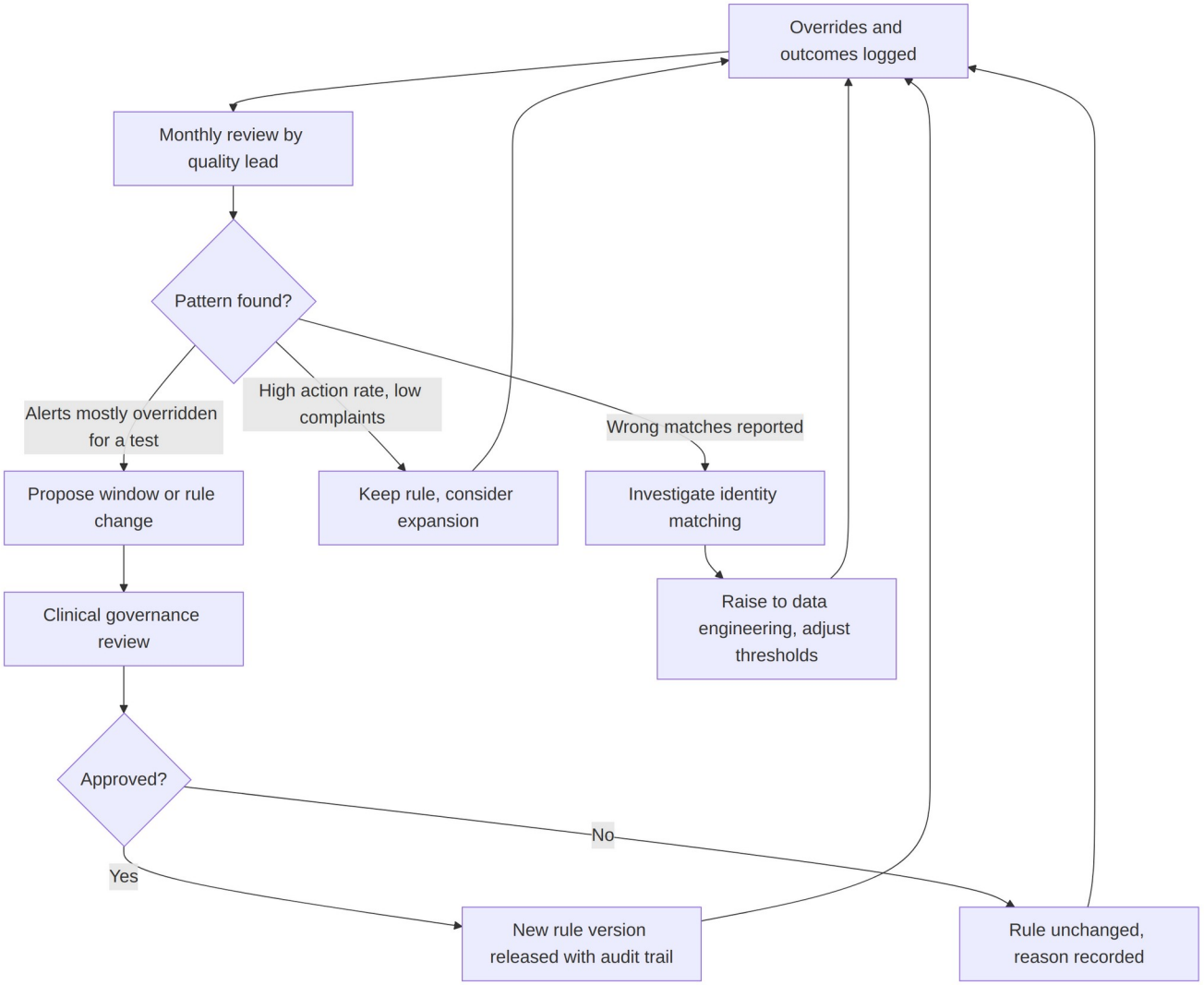
6. Data flow

Step	Data	From	To	Notes
1	Patient and order details (test code, priority)	EHR	DupeCheck	Minimum necessary fields only
2	Identity resolution	DupeCheck	Patient matching service	Returns match and confidence
3	Prior tests and results	Clinical exchange, local results, claims feed (v2)	DupeCheck	Claims show that a service was billed, not the result, and may lag
4	Test mapping	Reference tables (standard test codes)	DupeCheck	Maps local names to standard codes. Embedding based mapping is a gated v3 option
5	Rules and windows	Governed rules store	DupeCheck	Versioned, approved changes only
6	Alert card	DupeCheck	EHR	Reason, prior test, source, summary
7	Decision and reason	EHR	Audit log	Timestamp, user, outcome
8	Aggregates	Audit log	Analytics layer	De-identified by default

Data principles: synthetic data outside production, encryption in transit and at rest, role based access, audit of every view of external data.

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7. Override and governance flow



This loop keeps alerts calm and trusted. It is how alert fatigue is managed in practice.

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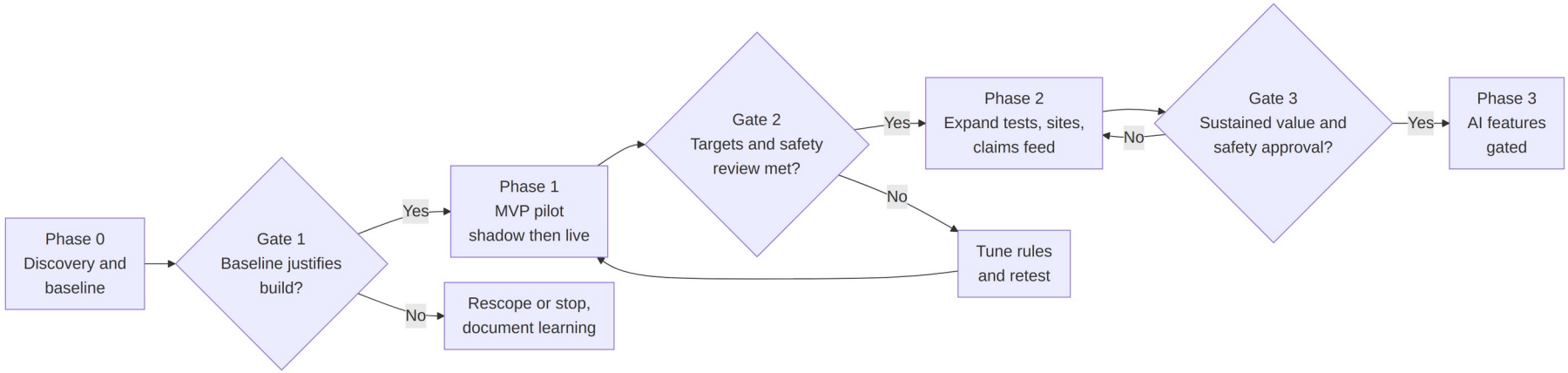
8. Roles and responsibilities across the flow

Step	Clinician	Quality lead	Analytics lead	Data engineering	Compliance	Product Owner
Place order and act on alert	R					
Review overrides and propose rule changes	C	R	C			A
Approve rule changes	C	R			C	A
Maintain data feeds and matching			C	R	C	A
Report avoided tests and cost		C	R	C		A
Audit access and use				C	R	A

R = responsible, A = accountable, C = consulted.

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9. Delivery flow: how the project itself moves



Phase detail

Phase	Key activities	Exit criteria
0 Discovery and baseline	Stakeholder interviews, workflow walkthroughs, data profiling, propose windows, confirm integration path, compliance review	Baseline duplicate rate measured, scope and rules signed off, decision gate passed

Phase	Key activities	Exit criteria
1 MVP pilot	Build rules engine and alert, shadow mode, limited live pilot, audit and insights	Action rate and false alert targets met, zero blocked urgent orders, safety review passed
2 Expansion	More tests and sites, claims feed, rule console, scheduled reports	Sustained value, low noise, governance running
3 Intelligence (gated)	NLP on report text, test mapping, smarter alerting	Validation and clinical safety approval

Ways of working: two week sprints, backlog in Jira, specs in Confluence, sprint reviews with clinicians, weekly pilot review in Phase 1, quarterly roadmap review.

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10. Traceability of this flow

Flow element	Linked BRD item	Linked PRD item
Section 3, alert at order time	BR-02, BR-03	US-01, US-03
Section 4, never block and urgent handling	BR-04	US-05
Section 4, identity first	Risk table	NFR, risk
Section 5 and 6, external and claims data	BR-09	US-02, F10
Section 7, governance	BR-06	US-09
Section 9, phases and gates	Timeline	Release plan